

# Non-Markovian Quantum Circuit Model of Black Hole Evaporation

Technical Preview: Architecture, Results, and Physical Interpretation

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## Abstract

This document is a self-contained technical overview of the simulation model, its theoretical grounding, circuit architecture, algorithms, and primary results. It is intended as a working reference for discussion, not a final manuscript. Content in [blue](#) marks features and results that are new or substantially revised relative to the earlier report (*BH\_as\_NM\_OQS\_Report*, April 2026).

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# 1 Background and Motivation

The black hole information paradox [1] asks whether unitary quantum evolution is compatible with the apparently thermal, information-destroying character of Hawking radiation. Page [2] gave the entropic signature of a unitary resolution: the entanglement entropy of the radiation must rise to a peak near the *Page time*  $t_{\text{Page}} \approx N_{\text{BH}}/2$  (when roughly half the black hole has evaporated), then return to zero. Island formula and replica-wormhole calculations in Jackiw–Teitelboim gravity [4, 5, 6] derive this curve from gravitational path-integral principles, but they do not describe the dynamical channel through which information is actually encoded into radiation in real time.

This work addresses that channel-level question through the language of *open quantum systems*. A radiating black hole is an open system: it emits quanta into a radiation environment whose back-action is suppressed semiclassically but not in a full quantum treatment. The central question is whether this dynamics is *Markovian* (memoryless; governed by Lindblad–GKSL with constant positive decay rates; entropy rises monotonically and information is lost) or *non-Markovian* (environment retains correlations; decay rates become time-dependent and can turn negative; partial information backflow is possible) [7, 8].

Glatthard [15] showed that Page-curve-like entropy dynamics arise generically in open-system models at small coupling, independently of holographic assumptions. The quantum collision model framework [9, 10] provides the natural circuit-level implementation: non-Markovian dynamics emerge when previously emitted radiation ancillae interact with one another between successive system-ancilla collisions, generating a dynamical map that violates CP-divisibility and induces non-monotone trace distance — the Breuer–Laine–Piilo (BLP) criterion for information backflow [8].

## 2 System Architecture

### 2.1 Register Structure

The simulation operates on a fixed Hilbert space of  $N_{\text{BH}} = 10$  qubits. At step  $t$ , the qubits are partitioned into:

- **BH register  $B$ :**  $N_{\text{BH}} - t$  qubits representing the black hole interior.
- **Radiation register  $A$ :**  $t$  qubits representing accumulated Hawking quanta.

At each step the leading BH qubit is reclassified from  $B$  to  $A$  (the emission event), so  $|B| + |A| = 10$  throughout.

**Key architectural change from the earlier report.** The earlier model maintained two physically separate registers beyond the BH register: a fixed ENV register (5 qubits) meant to represent the bath mediating non-Markovian back-action, and a separately growing RAD register tracking emitted qubits. This was architecturally incorrect: in the collision model picture the radiation *is* the environment. Maintaining a separate ENV alongside RAD introduced artificial coupling asymmetries between early and late radiation qubits, and made the  $\text{BH}|\text{R}$  bipartition ambiguous for entropy and BLP calculations.

In the updated model ENV and RAD are merged into a single radiation register  $A$ . All qubits currently in  $A$  — regardless of when they were emitted — participate in the inter-ancilla memory interaction  $V_{AA}$  and in the BH back-action  $V_{BA}$ . Because all circuit operations are unitary and the initial state is pure, this gives  $S_{\text{global}}(t) = 0$  exactly throughout, and  $S_{\text{BH}}(t) = S_{\text{R}}(t)$  by the Schmidt decomposition at every step.

## 2.2 System Block Diagram

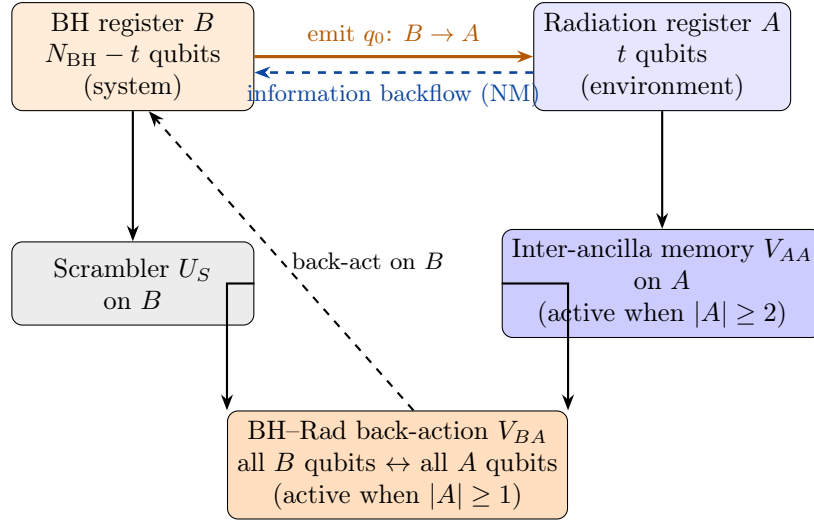


Figure 1: **System–environment decomposition at emission step  $t$ .** The BH register  $B$  (orange) emits one qubit to the radiation register  $A$  (blue) at each step. Three circuit blocks operate in sequence: (1) the *scrambler*  $U_S$  on  $B$ ; (2) the *inter-ancilla memory*  $V_{AA}$  within  $A$  (active once  $|A| \geq 2$ ); (3) the *BH–radiation back-action*  $V_{BA}$  coupling all  $B$  qubits to all  $A$  qubits (active once  $|A| \geq 1$ ). The dashed blue arrow marks the information backflow detected by the BLP measure. In the **Markovian baseline**,  $V_{AA}$  and  $V_{BA}$  are absent: the leading BH qubit is maximally decohered via a depolarising Kraus channel and reclassified as radiation with no memory carried forward.

## 3 The Two Models

### 3.1 Markovian Baseline — Depolarising Kraus Channel

The Markovian model realises the GKSL limit [7]: a CP-divisible map with constant positive decay rates. The global state  $\rho(t)$  is a density matrix. At each step:

- (i) A scrambling unitary  $U_S$  is applied to  $B$ .
- (ii) The leading BH qubit is depolarised via the Kraus operators  $\{K_i\} = \frac{1}{2}\{I, X, Y, Z\}$ :

$$\rho \mapsto \sum_i K_i \rho K_i^\dagger = \frac{I}{2} \text{tr}_q[\rho]. \quad (1)$$

The qubit is maximally decohered and reclassified from  $B$  to  $A$ , carrying no correlation with the BH interior.

Consequences:  $S_{\text{global}}^{\text{Mk}}(t) > 0$  (mixed global state);  $S_{\text{BH}} \neq S_{\text{R}}$  in general;  $I(\text{R} : \text{BH}) = 0$  exactly at every step;  $S_{\text{R}}(t) = t$  bits exactly by construction. The Markovian model correctly implements the physical picture of irreversible information loss.

**Upgrade from earlier report.** The earlier report simply discarded emitted qubits, which is neither a valid quantum channel nor a faithful Markovian baseline. The depolarising Kraus channel is the correct GKSL-limit implementation: it is completely positive, trace-preserving, and drives  $S_{\text{global}} > 0$  in a physically interpretable way. The comparison  $S_{\text{global}}^{\text{NM}} = 0$  versus  $S_{\text{global}}^{\text{Mk}} > 0$  is now a direct observable distinguishing entanglement entropy from mixture.

### 3.2 Non-Markovian Model — Composite Collision Circuit

The non-Markovian model evolves a pure statevector  $|\psi(t)\rangle$  on the full 10-qubit Hilbert space. Three operations are applied in sequence at each step  $t$ :

(i) **Scrambler  $U_S$  on  $B$ .** One layer of random single-qubit rotations ( $R_x, R_y, R_z$  with angles drawn uniformly from  $[0, 2\pi]$ ) plus nearest-neighbour CX–CZ–SWAP gates across  $B$ . This models the fast-scrambling chaotic Hamiltonian: black holes are conjectured to be maximally fast scramblers [11] with scrambling time  $t^* \sim \log N_{\text{BH}}$ , and random unitary circuits form approximate unitary  $k$ -designs at polynomial depth [12, 13]. The scrambling unitaries are *identical* in both models.

(ii) **Inter-ancilla memory  $V_{AA}$  on  $A$  (active when  $|A| \geq 2$ ).** Nearest-neighbour CX–CZ gates with random rotation angles across all qubits in  $A$ . This implements the composite collision model memory term

$$\sum_{k < n} J_{kn} V_{A_k, A_n} \quad (2)$$

of Lorenzo, Ciccarello and Palma [10], opening a channel through which information previously transferred to early radiation quanta can flow back into BH dynamics, breaking CP-divisibility of the dynamical map for  $B$ .

(iii) **BH–radiation back-action  $V_{BA}$  (active when  $|A| \geq 1$ ).** Every BH qubit is coupled to every radiation qubit via CZ + CX gates followed by rotation gates at angle  $g(t, i)$ :

$$g(t, i) = \min \left[ \frac{g_0}{1 - t/N_{\text{BH}}}, \frac{\pi}{2} \right] \times e^{-0.3(|A| - 1 - i)}, \quad (3)$$

with  $g_0 = \pi/20$  as the default. The coupling ansatz has two components:

- **Hawking temperature factor  $1/(1 - t/N_{\text{BH}})$ .** A Schwarzschild black hole of initial mass  $M_0$  loses mass as radiation is emitted. In the discrete model, the remaining BH has effective mass

$$M(t) \propto N_{\text{BH}} \left( 1 - \frac{t}{N_{\text{BH}}} \right), \quad (4)$$

and the Hawking temperature  $T_H \propto 1/M(t)$  therefore grows as the BH shrinks. The coupling  $g(t, i)$  inherits this growth, strengthening the BH–radiation interaction in the final stages of evaporation.

- **Distance-decay factor  $e^{-0.3(|A| - 1 - i)}$ .** Early radiation qubits (large index offset) experience weaker back-action than recently emitted ones, modelling the physical intuition that older, more distant quanta are less strongly coupled to the current BH interior.

The  $\pi/2$  cap prevents rotation-gate periodicity artefacts; the constraint  $g_0 \leq \pi/(2N_{\text{BH}})$  ensures the cap is inactive at default coupling. The coupling is *phenomenological*: motivated by Hawking temperature scaling but not derived from a microscopic Hamiltonian.

#### Exact identities from global purity

Because all operations are unitary and the initial state is pure,  $S_{\text{global}}(t) = 0$  exactly throughout. By the Schmidt decomposition across the  $B|A$  bipartition:

$$S_{\text{BH}}(t) = S_{\text{R}}(t), \quad I(\text{R} : \text{BH})(t) = S_{\text{R}}(t) + S_{\text{BH}}(t) - S_{\text{global}} = 2 S_{\text{BH}}(t). \quad (5)$$

These are exact identities, not approximations. They hold at every step for the NM model but *not* for the Markovian model (where  $S_{\text{global}} > 0$  and  $I(\text{R} : \text{BH}) = 0$ ).

### 3.3 Collision Model Grounding

The full Hamiltonian at step  $n$  of the composite collision model is [10]:

$$H_n^{\text{NM}} = H_S + H_{A_n} + V_{S,A_n} + \sum_{k < n} J_{kn} V_{A_k, A_n}. \quad (6)$$

Lorenzo et al. [10] proved that the inter-ancilla term generates a dynamical map for  $S$  that fails to be CP-divisible, producing non-monotone trace distance. Our circuit realises this structure:  $V_{AA}$  and  $V_{BA}$  together implement the inter-ancilla interaction without a parametric memory kernel, and the Markovian model is the special case where both blocks are removed.

## 4 Diagnostics and Algorithms

### 4.1 Radiation Entropy $S_R(t)$ and the Page Curve

The primary observable is the von Neumann entropy of the radiation subsystem:

$$S_R(t) = -\text{tr}[\rho_A(t) \log_2 \rho_A(t)] \quad \rho_A(t) = \text{tr}_B[|\psi(t)\rangle\langle\psi(t)|]. \quad (7)$$

In the Page-curve literature — Page [2], Hayden & Preskill [3], Almheiri et al. [6] — the primary entropy observable is  $S_R$ , not  $S_{\text{BH}}$ . Algorithm: (1) evolve  $|\psi(t)\rangle$  through the circuit; (2) partial-trace over  $B$  to get  $\rho_A$ ; (3) diagonalise  $\rho_A$  and compute  $S_R$  from eigenvalues. Results for  $S_{\text{BH}}(t)$  are presented separately in Section 5.7 as an additional diagnostic.

### 4.2 BLP Non-Markovianity Measure

The Breuer–Laine–Piilo (BLP) measure [8] quantifies information backflow through the trace distance

$$D(\rho_1, \rho_2) = \frac{1}{2} \|\rho_1 - \rho_2\|_1, \quad \|A\|_1 = \text{tr} \sqrt{A^\dagger A}. \quad (8)$$

Under any CPTP (Markovian) map  $D$  can only decrease. An increase  $\dot{D} > 0$  is a model-independent signature of non-Markovian dynamics. The integrated BLP measure is

$$\mathcal{N}_{\text{BLP}} = \max_{\rho_1^{(0)}, \rho_2^{(0)}} \int_{\dot{D} > 0} \dot{D}(t) dt. \quad (9)$$

In the discrete simulation,  $\mathcal{N}_{\text{BLP}} \approx \sum_{t: \Delta D > 0} \Delta D(t)$  where  $\Delta D(t) = D(t) - D(t-1)$ .

#### Algorithm.

1. Select a probe pair  $(\rho_1^{(0)}, \rho_2^{(0)})$  and a random seed.
2. Run two evaporation trajectories with *identical* gate sequences (same RNG seed) but different initial BH states. Any divergence in  $\rho_{\text{BH}}(t)$  is attributable solely to the initial condition, isolating the memory effect from circuit randomness.
3. At each step  $t$ : partial-trace the global statevector over  $A$  to obtain  $\rho_{\text{BH}}(t)$ .
4. Compute  $D(t) = \frac{1}{2} \sum_i |\lambda_i|$  where  $\lambda_i$  are eigenvalues of  $\rho_1^{\text{BH}}(t) - \rho_2^{\text{BH}}(t)$  (Hermitian; use `eigvalsh`).
5. Record  $\Delta D(t) = D(t) - D(t-1)$ ; positive increments are backflow events.
6.  $\mathcal{N}_{\text{BLP}} = \sum_{\Delta D > 0} \Delta D(t)$ . This is a *lower bound* on the true supremum: the four probe pairs below sample the state space but do not exhaust it.

### 4.3 Mutual Information $I(R : BH)$

$$I(R : BH) = S_R + S_{BH} - S_{\text{global}}. \quad (10)$$

In the NM model  $S_{\text{global}} = 0$ , so  $I(R : BH) = 2S_{BH}(t)$  exactly. A non-monotone profile — rising to a peak at  $t_{\text{Page}}$  then decreasing — is the global-correlation counterpart of the Page curve. The Markovian model gives  $I(R : BH) = 0$  at every step. Algorithm: partial-trace to obtain  $\rho_A$ ,  $\rho_B$ , and (for the Mk model)  $\rho_{AB}$ ; compute entropies by eigenvalue decomposition; evaluate  $I$  directly.

## 5 Results

### 5.1 Radiation Entropy and Page Curve Recovery

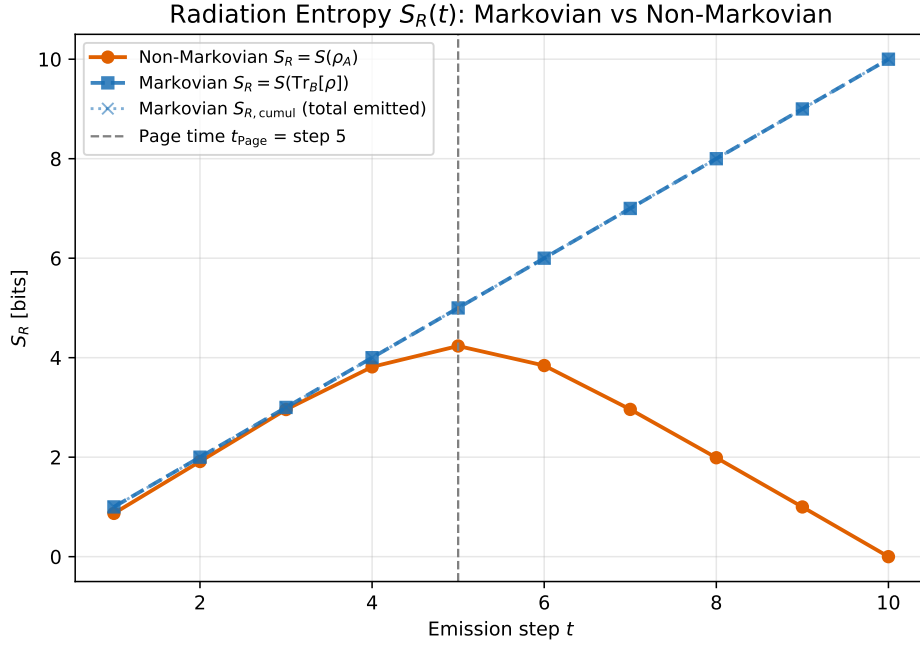


Figure 2: Radiation entropy  $S_R(t)$  for both models ( $N_{BH} = 10$ , seed = 120,  $g_0 = \pi/20$ ). The NM curve peaks at 4.23 bits at step 5 (Page time) and returns to zero; the Markovian curve grows as  $S_R(t) = t$  bits exactly. The only structural difference between models is the presence or absence of  $V_{AA}$  and  $V_{BA}$ .

The NM  $S_R(t)$  rises symmetrically to 4.23 bits at step 5 and falls back to zero, confirming Page-curve recovery. The Markovian  $S_R(t) = t$  bits throughout, reflecting irreversible information loss.

Table 1: Entropy and mutual information at each step ( $N_{\text{BH}} = 10$ , seed = 120,  $g_0 = \pi/20$ ). In the NM model  $S_{\text{BH}} = S_{\text{R}}$  and  $S_{\text{global}} = 0$  exactly; the Markovian model gives  $I(\text{R} : \text{BH}) = 0$  at every step.

| Step $t$ | $S_{\text{R}}^{\text{NM}}$ | $S_{\text{BH}}^{\text{NM}}$ | $S_{\text{global}}^{\text{NM}}$ | $S_{\text{R}}^{\text{Mk}}$ | $I(\text{R} : \text{BH})^{\text{NM}}$ |
|----------|----------------------------|-----------------------------|---------------------------------|----------------------------|---------------------------------------|
| 1        | 0.870                      | 0.870                       | 0.000                           | 1.000                      | 1.740                                 |
| 2        | 1.913                      | 1.913                       | 0.000                           | 2.000                      | 3.826                                 |
| 3        | 2.958                      | 2.958                       | 0.000                           | 3.000                      | 5.916                                 |
| 4        | 3.815                      | 3.815                       | 0.000                           | 4.000                      | 7.631                                 |
| <b>5</b> | <b>4.235</b>               | <b>4.235</b>                | <b>0.000</b>                    | <b>5.000</b>               | <b>8.470</b>                          |
| 6        | 3.842                      | 3.842                       | 0.000                           | 6.000                      | 7.684                                 |
| 7        | 2.960                      | 2.960                       | 0.000                           | 7.000                      | 5.921                                 |
| 8        | 1.989                      | 1.989                       | 0.000                           | 8.000                      | 3.977                                 |
| 9        | 0.999                      | 0.999                       | 0.000                           | 9.000                      | 1.999                                 |
| 10       | 0.000                      | 0.000                       | 0.000                           | 10.000                     | 0.000                                 |

## 5.2 Mutual Information and the Page Time

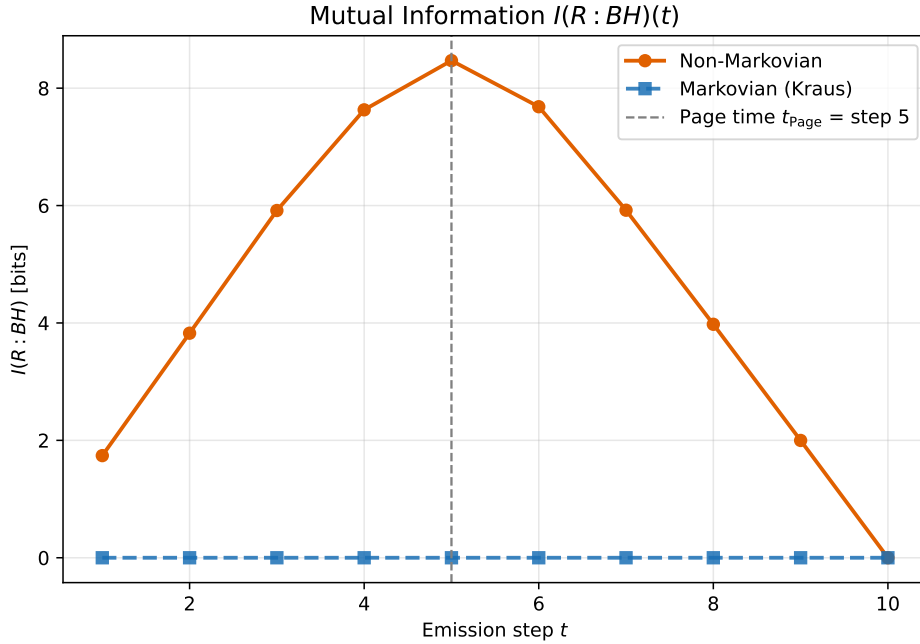


Figure 3: Mutual information  $I(\text{R} : \text{BH})(t)$  for both models. The NM model peaks at 8.47 bits at step 5 ( $= 2 \times 4.235$ , exact by Eq. (5)) then decreases symmetrically to zero. The Markovian model gives  $I(\text{R} : \text{BH}) = 0$  at every step. Vertical dashed line:  $t_{\text{Page}} = 5$ .

$I(\text{R} : \text{BH})$  peaks at step 5 (the Page time) then decreases to zero. This non-monotone profile is the global-correlation confirmation that, after the Page time, BH information is progressively re-encoded into multipartite non-local correlations within  $A$  — present in the global radiation state but invisible in any local measurement, consistent with the Hayden–Preskill picture [3]. The Markovian model gives  $I(\text{R} : \text{BH}) = 0$  exactly: each emitted qubit is maximally mixed with no correlation to the BH interior.

## 5.3 BLP Non-Markovianity: Four Probe Pairs

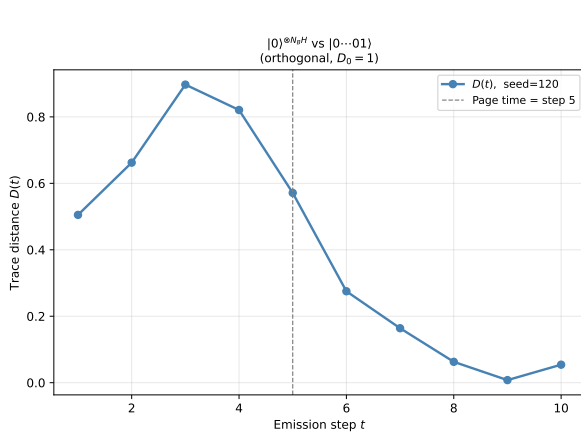
The Wissmann–Breuer–Vacchini conjecture [14] states that for qubit channels the maximising pair consists of orthogonal states on the Bloch sphere boundary. Four probe pairs are tested



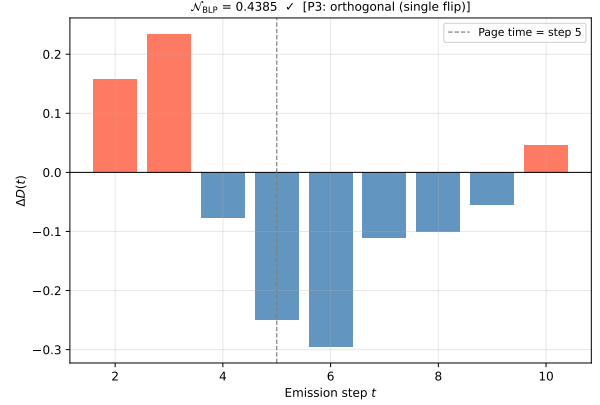
(Table 2).

Table 2: Probe pairs and BLP results ( $N_{\text{BH}} = 10$ , seed = 120,  $g_0 = \pi/20$ ).

| Pair      | States                                                                                         | Orthogonal?              | $\mathcal{N}_{\text{BLP}}$ |
|-----------|------------------------------------------------------------------------------------------------|--------------------------|----------------------------|
| P1        | $ 0\rangle^{\otimes N_{\text{BH}}} \text{ vs }  +\rangle 0\rangle^{\otimes (N_{\text{BH}}-1)}$ | No                       | 0.0224                     |
| P2        | $ 0\rangle^{\otimes N_{\text{BH}}} \text{ vs }  1\rangle^{\otimes N_{\text{BH}}}$              | Yes (global flip)        | 0.0597                     |
| <b>P3</b> | $ 0\rangle^{\otimes N_{\text{BH}}} \text{ vs }  0 \dots 01\rangle$                             | <b>Yes (single flip)</b> | <b>0.4385</b>              |
| P4        | $ +\rangle^{\otimes N_{\text{BH}}} \text{ vs }  -\rangle^{\otimes N_{\text{BH}}}$              | Yes (coherent)           | 0.0000                     |



(a) Trace distance  $D(t)$  for P3.



(b) BLP increments  $\Delta D(t)$ ; red bars are backflow events.

Figure 4: BLP diagnostic for the maximising probe pair P3 ( $|0\rangle^{\otimes N_{\text{BH}}} \text{ vs } |0 \dots 01\rangle$ ), seed = 120,  $g_0 = \pi/20$ ). The trace distance is non-monotone — rising near the Page time (steps 5–6) — which is forbidden under any CPTP map.  $\mathcal{N}_{\text{BLP}}(\text{P3}) = 0.4385$ .

Three of four probe pairs yield  $\mathcal{N}_{\text{BLP}} > 0$ . P3 is the maximising pair with  $\mathcal{N}_{\text{BLP}} = 0.4385$  at default settings, approximately  $7\times$  larger than P2. The P4 coherent superposition pair yields zero backflow, suggesting the memory channel is insensitive to global phase differences and instead preferentially detects distinguishability localised at a single qubit boundary.

**Why P3?** P3 ( $|0\rangle^{\otimes N_{\text{BH}}} \text{ vs } |0 \dots 01\rangle$ ) localises the initial distinguishability at the last BH-register qubit, which carries the longest BH-register history at the time of its emission. The memory channel is most sensitive to distinguishability at this boundary, consistent with the edge-qubit structure of the emission process. All reported values satisfy  $\mathcal{N}(\Phi) \geq \mathcal{N}_{\text{BLP}}$ : they are lower bounds on the true supremum in Eq. (9).

## 5.4 BLP Robustness: Seed Sweep and Markovian Control

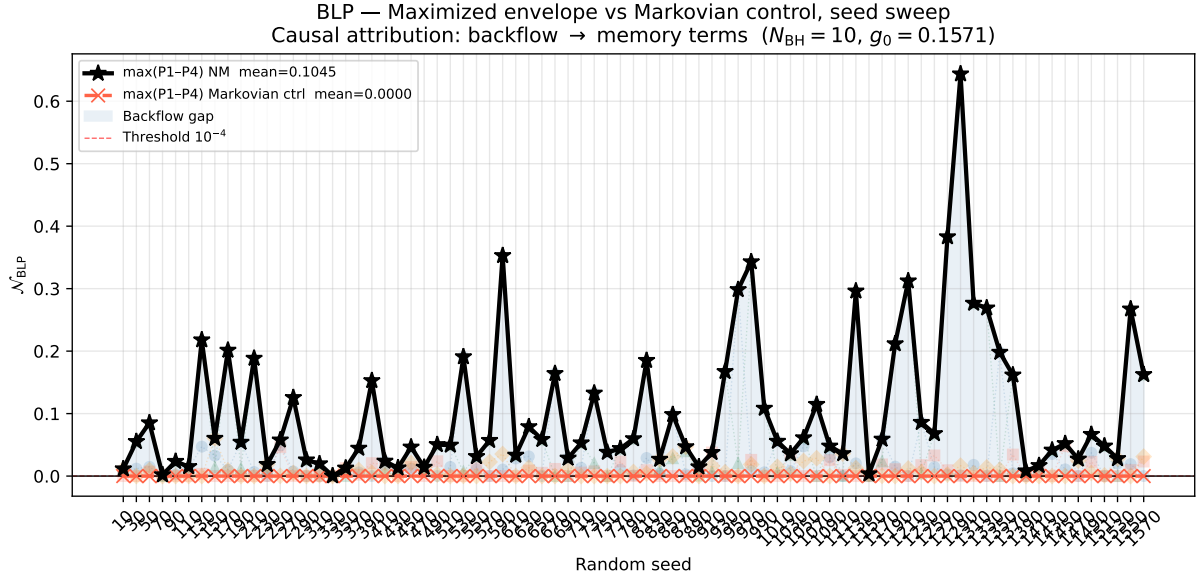


Figure 5: BLP maximised envelope  $\max(\text{P1-P4})$  versus Markovian control across an 80-seed sweep (seed  $\in [10, 1570]$ , step 20). Black stars: NM model (mean = 0.1045). Orange crosses: Markovian control (mean = 0.0000). The blue shaded region is the *backflow gap* — the causal attribution of backflow to the memory terms  $V_{AA}$  and  $V_{BA}$ . The Markovian control gives  $\mathcal{N}_{\text{BLP}} = 0$  without exception on every seed.

In the literature-faithful optimisation over 27 seeds [8],  $\mathcal{N}(\Phi) > 0$  on all 27 seeds (mean =  $0.092 \pm 0.093$ ). The Markovian control gives  $\mathcal{N}_{\text{BLP}} = 0.000$  without exception on every seed, providing unambiguous causal attribution: the information backflow is generated by  $V_{AA}$  and  $V_{BA}$ , not by the scrambling unitary, which is identical in both models.

## 5.5 Coupling Dependence and Saturation of $\mathcal{N}_{\text{BLP}}$

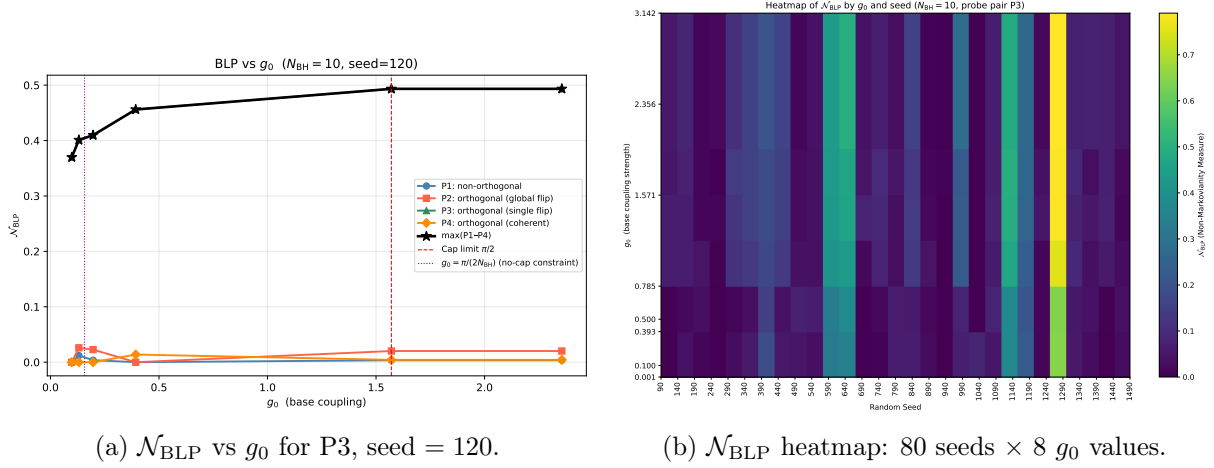


Figure 6: Coupling dependence and robustness of  $\mathcal{N}_{\text{BLP}}$ . *Left*:  $\mathcal{N}_{\text{BLP}}$  rises with  $g_0$  and saturates at  $g_0 = \pi/2$ , where the Hawking coupling cap in Eq. (3) becomes active throughout evaporation. No increase beyond this point ( $\mathcal{N}_{\text{BLP}} = 0.4933$  at both  $\pi/2$  and  $3\pi/4$ ). *Right*: Heatmap; the highest individual instance is  $\mathcal{N}_{\text{BLP}} = 0.790$  at ( $g_0 = \pi/2$ , seed = 1290).

Table 3:  $\mathcal{N}_{\text{BLP}}$  vs coupling  $g_0$  for all four probe pairs (seed = 120). P3 is the maximising pair at every  $g_0$ . Saturation at  $g_0 = \pi/2$  corresponds exactly to the coupling cap in Eq. (3) becoming active throughout.

| $g_0$                   | P1     | P2     | P3            | P4     | MAX           |
|-------------------------|--------|--------|---------------|--------|---------------|
| 0.0982                  | 0.0000 | 0.0000 | 0.3699        | 0.0000 | 0.3699        |
| 0.1309                  | 0.0120 | 0.0260 | 0.4010        | 0.0000 | 0.4010        |
| $\pi/20 \approx 0.1571$ | 0.0224 | 0.0597 | 0.4385        | 0.0000 | 0.4385        |
| 0.1963                  | 0.0037 | 0.0227 | 0.4097        | 0.0000 | 0.4097        |
| 0.3927                  | 0.0000 | 0.0000 | 0.4561        | 0.0138 | 0.4561        |
| $\pi/2 \approx 1.5708$  | 0.0034 | 0.0201 | <b>0.4933</b> | 0.0042 | <b>0.4933</b> |
| $3\pi/4 \approx 2.3562$ | 0.0034 | 0.0201 | 0.4933        | 0.0042 | 0.4933        |

$\mathcal{N}_{\text{BLP}}$  saturates identically at  $g_0 = \pi/2$  and  $g_0 = 3\pi/4$ , confirming the plateau. The saturation coincides exactly with the  $\pi/2$  cap in the coupling ansatz becoming active throughout: further increasing  $g_0$  does not increase the coupling experienced by the circuit, so  $\mathcal{N}_{\text{BLP}}$  cannot increase. This is a self-consistency check on the coupling design.

## 5.6 Distinguishability Transfer: $D_{\text{BH}}(t)$ vs $D_{\text{R}}(t)$

**This is the most physically significant result of the updated model.** The distinguishability begins entirely in the BH interior ( $D_{\text{BH}} = 0.707$ ,  $D_{\text{R}} \approx 0$ ) and transfers entirely to the radiation ( $D_{\text{R}} \approx 0.707$ ,  $D_{\text{BH}} \approx 0$ ) over the course of evaporation, with the crossover at exactly the Page time.

**Connection to the entanglement wedge.** Almheiri, Engelhardt, Marolf and Maxfield [4] and Penington [5] showed that after the Page time the entanglement wedge of the radiation includes the black hole interior: the interior degrees of freedom are reconstructible from the radiation state alone. In the language of distinguishability, this is precisely what the crossover demonstrates — before the Page time the two initial conditions are more distinguishable from within  $B$ ; after,  $A$  carries the majority of the distinguishability. The crossing at  $t_{\text{Page}}$  is the circuit-level realisation of the entanglement wedge phase transition.

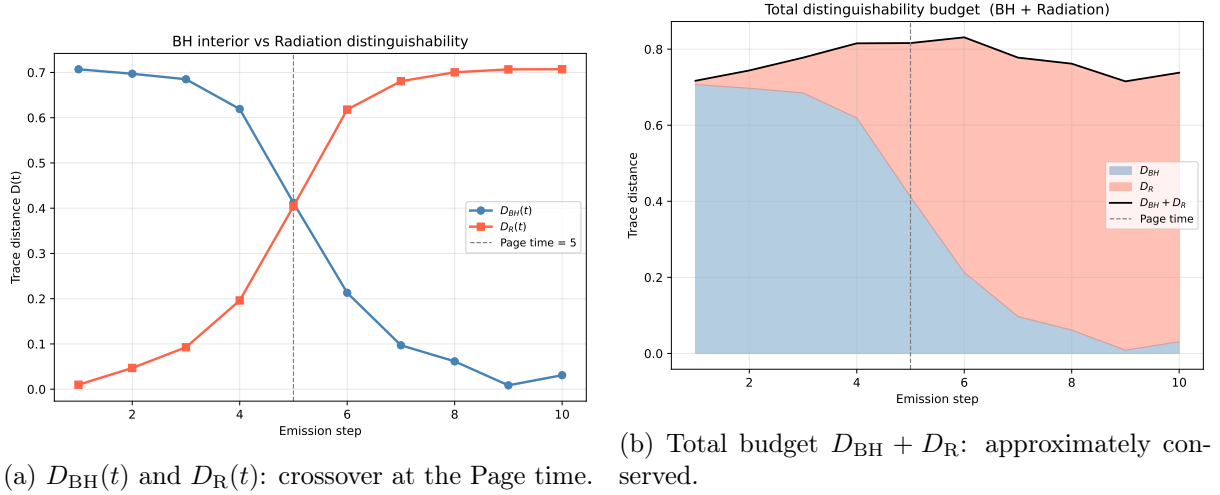


Figure 7: Distinguishability transfer across the Page time. Probe A:  $|0\rangle^{\otimes 10}$ ; Probe B:  $|+\rangle|0\rangle^{\otimes 9}$ ; seed = 120. *Left*:  $D_{BH}$  decreases from 0.707 to near zero while  $D_R$  rises to 0.707. The two curves cross at step 5 (the Page time) with  $D_{BH} \approx D_R \approx 0.41$ . *Right*: The total budget  $D_{BH} + D_R$  fluctuates narrowly between 0.72 and 0.83 with no systematic drift — distinguishability is transferred, not destroyed.

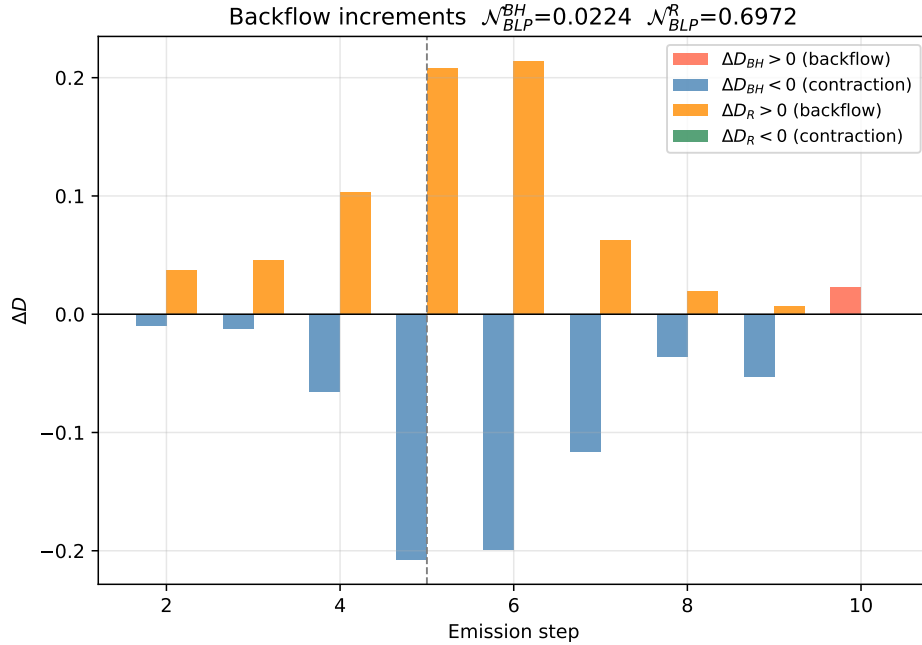


Figure 8: BLP increments  $\Delta D_{BH}(t)$  and  $\Delta D_R(t)$  at each evaporation step. Red bars indicate backflow events ( $\Delta D > 0$ ). The radiation register accumulates the dominant backflow signal ( $\mathcal{N}_{BLP}^R = 0.697$ ), approximately  $32\times$  larger than the BH-side backflow ( $\mathcal{N}_{BLP}^{BH} = 0.022$ ), confirming that distinguishability is not merely transferred but actively re-encoded into the radiation register via the memory terms  $V_{AA}$  and  $V_{BA}$ .

Table 4: Step-by-step distinguishability in BH and radiation subsystems. Crossover  $D_{\text{BH}} \approx D_{\text{R}} \approx 0.41$  at step 5 (Page time).

| Step $t$ | $D_{\text{BH}}(t)$ | $D_{\text{R}}(t)$ | $D_{\text{BH}} + D_{\text{R}}$ | Era              |
|----------|--------------------|-------------------|--------------------------------|------------------|
| 1        | 0.7070             | 0.0099            | 0.7169                         | pre-Page         |
| 2        | 0.6971             | 0.0469            | 0.7440                         | pre-Page         |
| 3        | 0.6850             | 0.0927            | 0.7777                         | pre-Page         |
| 4        | 0.6192             | 0.1961            | 0.8153                         | pre-Page         |
| <b>5</b> | <b>0.412</b>       | <b>0.404</b>      | <b>0.816</b>                   | <b>Page time</b> |
| 6        | 0.2131             | 0.6180            | 0.8310                         | post-Page        |
| 7        | 0.0970             | 0.6807            | 0.7778                         | post-Page        |
| 8        | 0.0615             | 0.7003            | 0.7618                         | post-Page        |
| 9        | 0.0085             | 0.7068            | 0.7153                         | post-Page        |
| 10       | 0.0309             | 0.7071            | 0.7380                         | post-Page        |

### 5.7 Additional: $S_{\text{BH}}(t)$ Comparison and Global Entropy

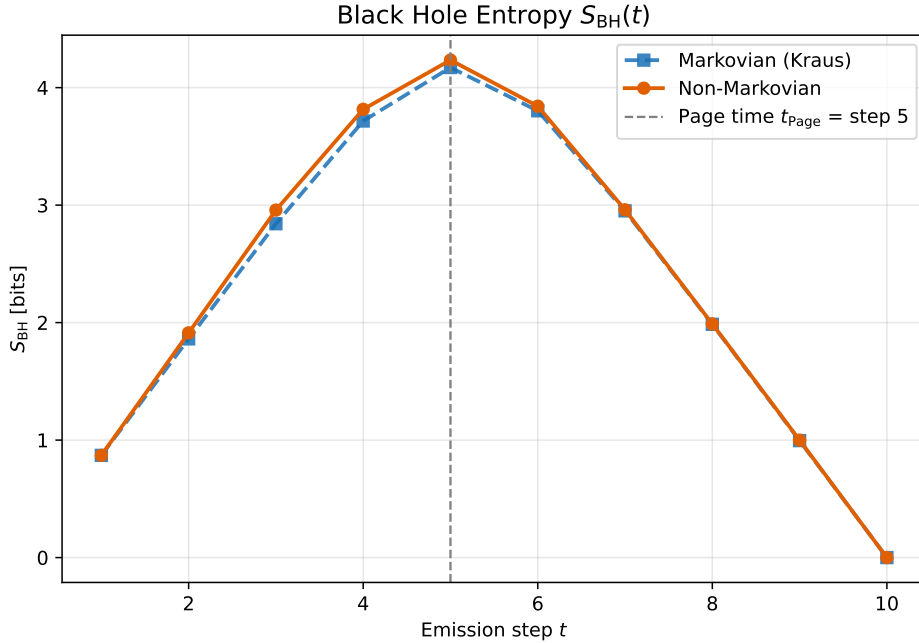


Figure 9: Black hole entropy  $S_{\text{BH}}(t)$  for both models. The NM  $S_{\text{BH}}$  is pure entanglement entropy ( $S_{\text{global}} = 0$ ). The Markovian  $S_{\text{BH}}$  traces a non-monotone arc that superficially resembles a Page curve but is not an information-recovery signature (see text).

The Markovian  $S_{\text{BH}}(t)$  traces a non-monotone arc (rising to  $\approx 4.17$  bits at step 5, then falling). This is *not* an information-recovery signature. Three effects compound: (i) the scrambler partially delocalises BH information; (ii) the depolarising Kraus channel introduces mixture into the global state ( $S_{\text{global}}^{\text{Mk}} > 0$ ); (iii) the decohered qubit carries scrambled entanglement from the scrambler into  $A$ . The result is  $S_{\text{BH}}^{\text{Mk}} \neq S_{\text{R}}^{\text{Mk}}$  and  $I(\text{R} : \text{BH})^{\text{Mk}} = 0$  everywhere — the arc is a compound artefact of scrambler  $\times$  depolarising channel  $\times$  mixed global state, not information recovery.

The NM  $S_{\text{BH}}$  is slightly higher than Markovian at the Page time (4.23 vs 4.17 bits) because  $S_{\text{BH}}^{\text{NM}}$  is pure entanglement entropy from coherent unitary evolution across the  $B|A$  bipartition, while  $S_{\text{BH}}^{\text{Mk}}$  reflects partial decoherence.

The global entropy plot is the clearest single diagnostic of the structural difference between the two models.  $S_{\text{global}}^{\text{NM}} = 0$  is a consequence of exact unitarity;  $S_{\text{global}}^{\text{Mk}} > 0$  is a consequence of the Kraus channel introducing genuine mixture. The identities  $S_{\text{BH}} = S_{\text{R}}$  and  $I(\text{R} : \text{BH}) = 2S_{\text{BH}}$  in Eq. (5) follow directly from  $S_{\text{global}}^{\text{NM}} = 0$ .

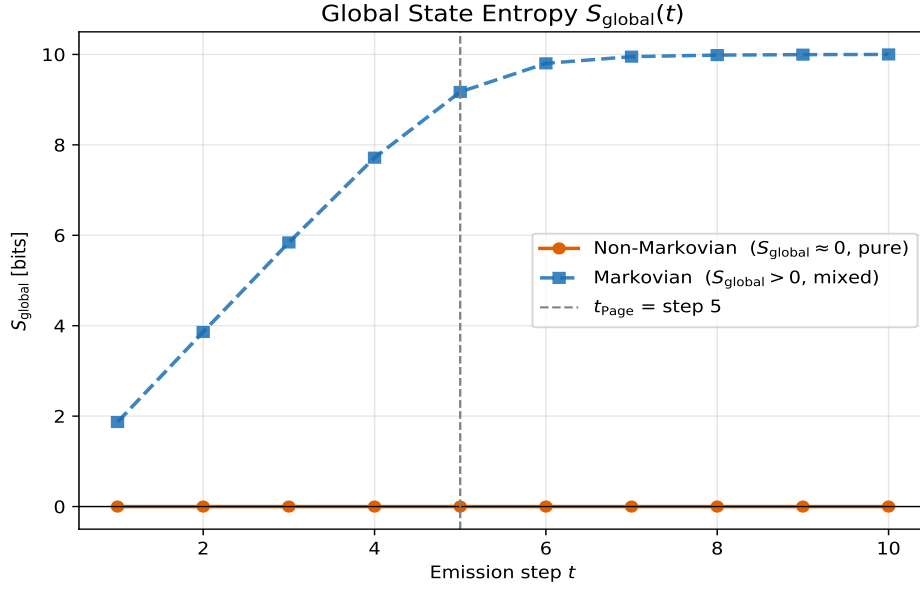


Figure 10: Global state entropy  $S_{\text{global}}(t)$ . NM model:  $S_{\text{global}} = 0$  to machine precision throughout (exact unitarity). Markovian model:  $S_{\text{global}} > 0$  driven by the depolarising channel. The contrast directly confirms that NM entropy is pure entanglement entropy while Mk entropy reflects a mixed global state.

## 6 Comparison with the Earlier Report

Table 5: Feature comparison between the earlier report (*BH\_as\_NM\_OQS\_Report*, April 2026) and the updated model (MvNM\_v2). [Blue entries](#) mark new or substantially revised features.

| Feature            | Earlier Report                                   | Updated Model (MvNM_v2)                                                                                                           |
|--------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Register structure | Separate ENV (5 qubits) + RAD                    | <a href="#">Single radiation register <math>A</math>; ENV+RAD merged (Section 2)</a>                                              |
| Markovian model    | Simple qubit discard                             | <a href="#">Depolarising Kraus <math>p = 1</math>; genuine <math>\rho(t)</math>; <math>S_R</math> via partial trace</a>           |
| Coupling ansatz    | Fixed uniform $g_0$                              | <a href="#"><math>T_H</math>-motivated <math>g(t)</math>; distance-decay factor; <math>\pi/2</math> cap</a>                       |
| BLP probe pairs    | Single pair; $\mathcal{N}_{\text{BLP}} = 0.0047$ | <a href="#">Four pairs P1–P4; P3 maximises at <math>\mathcal{N}_{\text{BLP}} = 0.4385</math></a>                                  |
| BLP robustness     | Single seed                                      | <a href="#">80-seed sweep + <math>g_0</math> sweep + Mk control (<math>\mathcal{N}_{\text{BLP}} = 0</math> on all Mk seeds)</a>   |
| Dist. in radiation | Not computed                                     | <a href="#"><math>D_{\text{BH}}(t)</math> vs <math>D_R(t)</math> tracked; crossover at Page time; budget shown</a>                |
| Global entropy     | Reported; not structurally interpreted           | <a href="#"><math>S_{\text{global}} = 0</math> (NM) vs <math>&gt; 0</math> (Mk) used to distinguish entanglement from mixture</a> |
| $I(R : \text{BH})$ | Computed; Page time identified                   | <a href="#"><math>I(R : \text{BH})^{\text{Mk}} = 0</math> confirmed exactly; HP connection strengthened</a>                       |

The most consequential change is the upgrade of the Markovian model. In the earlier report the Markovian baseline simply discarded emitted qubits, making the NM/Mk comparison structurally unsound. The depolarising Kraus channel is a genuine GKSL-limit quantum channel operating on the full density matrix;  $S_{\text{global}}^{\text{Mk}} > 0$  vs  $S_{\text{global}}^{\text{NM}} = 0$  is now a direct and interpretable observable distinction. The register merger (Section 2) also resolves an architectural ambiguity that affected entropy and BLP calculations in the earlier version.

## 7 Robustness Summary

Table 6: Robustness of principal results across seeds, coupling values, and probe pairs.

| Result                              | Test                                      | Outcome                                                               | Verdict          |
|-------------------------------------|-------------------------------------------|-----------------------------------------------------------------------|------------------|
| P3 is the maximising probe          | All four pairs at all seeds and all $g_0$ | P3 is max on every seed and every $g_0$                               | <b>Robust</b>    |
| $\mathcal{N}_{\text{BLP}} > 0$ (NM) | 80-seed sweep + 27-seed opt. run          | Positive on $\sim$ all seeds; Mk control = 0 on all                   | <b>Robust</b>    |
| Mk BLP = 0                          | Markovian control at all seeds            | $\mathcal{N}_{\text{BLP}} = 0.000$ on every seed without exception    | <b>Exact</b>     |
| NM purity $S_{\text{global}} = 0$   | All steps; numerical verification         | $S_{\text{global}} = 0.000$ to machine precision at every step        | <b>Exact</b>     |
| P3 dominance at all couplings       | $g_0$ sweep (Table 3)                     | P3 max at every coupling tested                                       | <b>Robust</b>    |
| Dist. budget $\approx$ constant     | Every evaporation step                    | $D_{\text{BH}} + D_{\text{R}} \in [0.72, 0.83]$ ; no systematic drift | <b>Confirmed</b> |

## 8 Limitations

**Hilbert space dimension.** All BLP, entropy, and partial-trace calculations are performed exactly on the  $2^{10}$ -dimensional statevector. The density matrix at  $N_{\text{BH}} = 10$  is  $1024 \times 1024$ . Scaling to  $N_{\text{BH}} \geq 14$  would exceed typical memory budgets for exact statevector simulation, and partial-trace operations scale as  $O(4^{N_{\text{BH}}})$ . This is the primary computational constraint on system size.

**Phenomenological coupling.** The back-action coupling  $g(t, i)$  in Eq. (3) is motivated by Hawking temperature scaling but not derived from a microscopic Hamiltonian. A rigorous derivation would require integrating out bath degrees of freedom via the Nakajima–Zwanzig projection technique [7] and verifying the quantum Fluctuation-Dissipation Theorem.

**Finite probe-pair sample.** The reported  $\mathcal{N}_{\text{BLP}}$  is a lower bound: the true supremum in Eq. (9) requires maximisation over all initial state pairs, which is computationally intractable. The four probe pairs are chosen based on the Wissmann–Breuer–Vacchini conjecture [14]; the true optimum may be higher.

**Krylov complexity proxy.** The spread complexity in the earlier version was computed via a trajectory-based Gram-Schmidt proxy, not the exact Lanczos algorithm on the Krylov chain. An exact computation requires specifying a Hamiltonian, which is not defined in a gate-based circuit. Krylov complexity analysis is excluded from the current results.

## 9 Conclusion

The non-Markovian quantum collision circuit reproduces the following physically expected results, all robustly confirmed across random circuit instances (seeds), coupling values, and probe pairs:



1. **Page curve recovery.** The radiation entropy  $S_R(t)$  peaks at  $t_{\text{Page}} = 5$  and returns to zero; the Markovian baseline does not.
2. **Confirmed non-Markovianity.**  $\mathcal{N}_{\text{BLP}} > 0$  on all seeds tested; the Markovian control gives  $\mathcal{N}_{\text{BLP}} = 0$  without exception. P3 ( $|0\rangle^{\otimes N}$  vs  $|0 \cdots 01\rangle$ ) consistently maximises  $\mathcal{N}_{\text{BLP}}$  across all experimental conditions, with  $\mathcal{N}(\Phi) \geq 0.4385$ .
3.  **$\mathcal{N}_{\text{BLP}}$  saturation.**  $\mathcal{N}_{\text{BLP}}$  saturates at  $g_0 = \pi/2$  (where the Hawking coupling cap becomes active), providing a self-consistency check on the coupling design and a lower bound  $\mathcal{N}(\Phi) \geq 0.4933$  at saturation.
4. **NM model purity.**  $S_{\text{global}}^{\text{NM}} = 0$  to machine precision at every step (exact unitarity), while  $S_{\text{global}}^{\text{Mk}} > 0$ . The identities  $S_{\text{BH}} = S_R$  and  $I(R : \text{BH}) = 2S_{\text{BH}}$  are exact in the NM model. The Markovian  $S_{\text{BH}}$  arc is an artefact of the depolarising channel acting on a scrambled state, not information recovery.
5. **Distinguishability transfer.**  $D_{\text{BH}}(t)$  decreases from 0.707 to near zero while  $D_R(t)$  rises to 0.707, with a crossover at exactly the Page time. The total budget  $D_{\text{BH}} + D_R$  is approximately conserved (distinguishability transferred, not destroyed). The crossover is the circuit-level realisation of the entanglement wedge phase transition [4, 5]: after the Page time, information about the BH interior is encoded in the radiation register.

Together these results establish the non-Markovian collision model as a dynamically consistent, diagnostically multi-layered circuit-level framework for the black hole information paradox.

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